www.aethom00.github.io
www.linkedin.com/in/aethom

aethom@umich.edu
+1 231 492 8156
1 River Court, NJ, USA

Ashton E. Thomas

Education

University of Michigan - Rackham Graduate School 08/2025 – 05/2026
Major - MSE Computer Engineering — Concentration in ML/CV *Ann Arbor, MI, USA*

University of Michigan College of Engineering 08/2022 – 05/2025
Major - BSE Computer Science, GPA - 3.81 / 4.0 *Ann Arbor, MI, USA*
Relevant coursework: AR/VR, Computer Organization, Computer Vision, Cryptography, Data Analytics Data Structures & Algorithms, Operating Systems, Quantum Computing, Web Development, Web Systems

Experience

Amazon 09/2024 – Present
Software Development Engineer *Boston, MA & New York City, USA*
- Partnered with a team of more than 20 engineers to design, test, and optimize Alexa devices, enhancing functionality and elevating user experience.
- Independently developed and implemented new features for Alexa devices, optimizing performance and user experience with the use of React Native, Kotlin, TypeScript, and related technologies.
- Worked in a team of 10 engineers to manage Amazon company-wide employee data regarding 100,000s employees, including the collection, processing, and inferences used by corporate executives at the top of Amazon.

Ground Vehicle Systems Center (SEC) 05/2024 – 08/2024
Software Engineer *Warren, MI, USA*
- Leveraged MagicDraw & Excel to design databases for Jira tickets and hardware, leading to improved efficiency.
- Created Python scripts to parse large csvs with 1000s of datapoints to update integrated networks in Jira.

Madi Taylor Photo 06/2021 – 07/2024
Full Stack Developer *Traverse City, MI, USA*
- Developed and maintained the corporate website, crafting a cohesive user interface with HTML, CSS, and JS.
- Implemented robust back-end payment solutions and form validation to streamline user transactions.

Research

Polk Lab — *University of Michigan* 09/2024 – Present
Computational Neuroscience Research Assistant *Ann Arbor, MI, USA*
- Collaborated with a PhD student to develop a machine learning model aimed at mimicking neural distinctiveness of the human brain, while also enabling the separation of background noise from spoken language.
- Worked alongside Prof. Thad Polk to develop a script that processes CSV files with 1000s of columns, utilizing wildcard parsing to dynamically filter columns. The script includes a node hierarchy for efficient column interrelation and supports operations such as adding, removing, printing, and row filtration.

Projects

Geoguessr AI, Computer Vision 2023 – 2024
- Designed and implemented a modified ResNet-50 architecture for geographic location identification from images.
- Fine-tuned the model with 61k images, addressing lighting and seasonal challenges whilst achieving ∼90% accuracy.

Google Search Engine, Web Systems 2024
- Engineered a scalable search engine leveraging a segmented inverted index implemented with MapReduce.
- Integrated tf-idf for text analysis and PageRank for link analysis to improve the relevance of search results produced via a REST API.

Instagram Clone, Web Systems 2024
- Created a server-side dynamic version of Instagram with Flask and SQL relational databases, enabling features like user authentication, content management, and interactions.
- Implemented a client-side dynamic version of Instagram using JavaScript, React, and REST APIs, introducing seamless content updates, infinite scroll, and double-tap to like functionality.

Study Group Coordinator, Quantum Computing 2024
- Designed and developed a Study Group Scheduler with Quantum algorithms leveraging Grover's algorithm for efficient group formation under CNF (Conjunctive Normal Form) constraints.
- Created and implemented Bitflip and Phase Oracles to translate CNF constraints into quantum operations.

Technical Skills

AJAX, ARM, ARCGIS, C, C#, C++, CSS, Excel, Flask, Git, HTML, Javascript, Jinja, Jira, JSON, Keras, Kotlin, Latex, MagicDraw, Makefile, Matlab, NumPy, Python, PyTorch, Qiskit, React/Native, SQL, Typescript, Unreal Engine, VSCode